

The Focusing Cubic NLS Soliton: Exact Hessian Spectrum and Pöschl–Teller Factorization

Route-A structural certificate HCS-C221

Abstract

We close a convention-locked one-dimensional theorem for the focusing cubic nonlinear Schrödinger equation. The sech standing wave has exact mass, Hamiltonian, action and Vakhitov–Kolokolov (VK) slope. Its two real Hessians have the complete Pöschl–Teller discrete spectrum, essential threshold, kernels and Morse index. The zero-frequency and defocusing faces are separated, and no finite-box spectrum or arithmetic interpretation is used. The variational scaling and explicit eigenfunctions are included. The ladder factorization, boundary ledger, independent evidence counts and strict Route-A stop are included.

Keywords: nonlinear Schrödinger; soliton; Hessian; Pöschl–Teller; Morse index; Route A.

1 Equation and standing wave

We study

$$i\psi_t + \psi_{xx} + 2|\psi|^2\psi = 0, \quad \psi(t, x) = e^{i\omega t}Q_\omega(x), \quad \omega > 0, \quad (1)$$

on $\mathbb{R}$. Direct differentiation of $Q_\omega(x) = \sqrt{\omega} \operatorname{sech}(\sqrt{\omega}x)$ gives

$$-Q''_\omega + \omega Q_\omega - 2Q_\omega^3 = 0. \quad (2)$$

For

$$M(u) = \int_{\mathbb{R}} |u|^2 dx, \quad H(u) = \frac{1}{2} \int_{\mathbb{R}} |u_x|^2 dx - \frac{1}{2} \int_{\mathbb{R}} |u|^4 dx, \quad S_\omega = H + \frac{\omega}{2} M,$$

the substitution $y = \sqrt{\omega}x$, $z = \tanh y$ yields

$$M(Q_\omega) = 2\sqrt{\omega}, \quad \|Q'_\omega\|_2^2 = \frac{2}{3}\omega^{3/2}, \quad \|Q_\omega\|_4^4 = \frac{4}{3}\omega^{3/2}, \quad (3)$$

and hence

$$H(Q_\omega) = -\frac{1}{3}\omega^{3/2}, \quad S_\omega(Q_\omega) = \frac{2}{3}\omega^{3/2}, \quad \frac{d}{d\omega}M(Q_\omega) = \omega^{-1/2} > 0. \quad (4)$$

2 The complete Hessian spectrum

The real and imaginary second variations are the self-adjoint operators

$$L_+ = -\partial_x^2 + \omega - 6\omega \operatorname{sech}^2(\sqrt{\omega}x), \quad L_- = -\partial_x^2 + \omega - 2\omega \operatorname{sech}^2(\sqrt{\omega}x). \quad (5)$$

The theorem is

$$\sigma_{\text{ess}}(L_+) = \sigma_{\text{ess}}(L_-) = [\omega, \infty), \quad \sigma_{\text{disc}}(L_+) = \{-3\omega, 0\}, \quad \sigma_{\text{disc}}(L_-) = \{0\}. \quad (6)$$

The -3ω eigenfunction is $\operatorname{sech}^2(\sqrt{\omega}x)$; the zero eigenfunctions are Q'_ω for L_+ and Q_ω for L_- . All listed eigenvalues are simple. Thus L_+ has Morse index one and kernel $\operatorname{span}\{Q'_\omega\}$, while $L_- \geq 0$ and has kernel $\operatorname{span}\{Q_\omega\}$.

To see the signs without a box discretization, put $y = \sqrt{\omega}x$ and

$$A_\ell = \partial_y + \ell \tanh y, \quad A_\ell^* = -\partial_y + \ell \tanh y.$$

Multiplication gives

$$-\partial_y^2 + 1 - 6 \operatorname{sech}^2 y = A_2^* A_2 - 3, \quad -\partial_y^2 + 1 - 2 \operatorname{sech}^2 y = A_1^* A_1. \quad (7)$$

The $\ell = 1, 2$ ladder has exactly the bound states displayed in (6); the free remainder starts at one, which rescales to the threshold ω .

3 Boundaries and evidence

At $\omega \downarrow 0$, the profile and discrete eigenvalues collapse to the essential threshold. Reversing the cubic sign has no nonzero bright H^1 sech branch. A periodic domain gives elliptic/cnoidal waves, and dimensions $d \geq 2$ have different criticality; neither is silently included. The executable receipt contains 15 profile, 3 integral, 15 spectral and 15 factorization rows. An independent checker passes 497 assertions, SymPy passes 19 identities, replay is byte exact, and 17 repaired/stale-hash hostile mutations are rejected. These rows are regression evidence only; they do not prove the continuum statement by sampling. In particular, no full nonlinear orbital-stability theorem is asserted.

4 Strict Route-A boundary

The continuum NLS coefficients carry no intrinsic rational-prime or prime-power clock, and the standing-wave symmetry supplies only a weak relative orbit hint. The strict tuple is

$$(A0, A1, A2, A3, A4) = (AO_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).$$

The result is `ROUTE_A_REJECTED`; Route B is disabled. The final entry records only a natural Hamiltonian quantization hint, not a Hilbert–Pólya operator. Hessian traces and continuous spectra are not dynamical zeta functions, Euler factors or target divisors. The locked scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. The source baseline lock for this round is `86c7bb8a39cdd1b8e941e45833b068170ca06287`.

The source-attributed references are Zakharov–Shabat [ZS72], Weinstein [Weinstein85], and the historical Pöschl–Teller solvable-potential/ladder source [PT33]; no priority claim is made.

Revision focus. Round 2 adds factorization (7), all scope faces, independent evidence counts, provenance and the strict Route-A stop.

References

- [1] V. E. Zakharov and A. B. Shabat, “Exact theory of two-dimensional self-focusing and one-dimensional self-modulation of waves in nonlinear media,” *Soviet Physics JETP* 34 (1972), 62–69. Official record.
- [2] M. I. Weinstein, “Modulational Stability of Ground States of Nonlinear Schrödinger Equations,” *SIAM J. Math. Anal.* 16 (1985), 472–491. DOI: 10.1137/0516034.
- [3] G. Pöschl and E. Teller, “Bemerkungen zur Quantenmechanik des anharmonischen Oszillators,” *Zeitschrift für Physik* 83 (1933), 143–151. DOI: 10.1007/BF01331132. Cited only as the historical solvable Pöschl–Teller potential/ladder source.

Declarations

Data and code. Synthetic exact probes and package-local deterministic code accompany the manuscript; no training data are used. **Ethics/funding.** No human or animal data; no external funding reported. **AI disclosure.** An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.